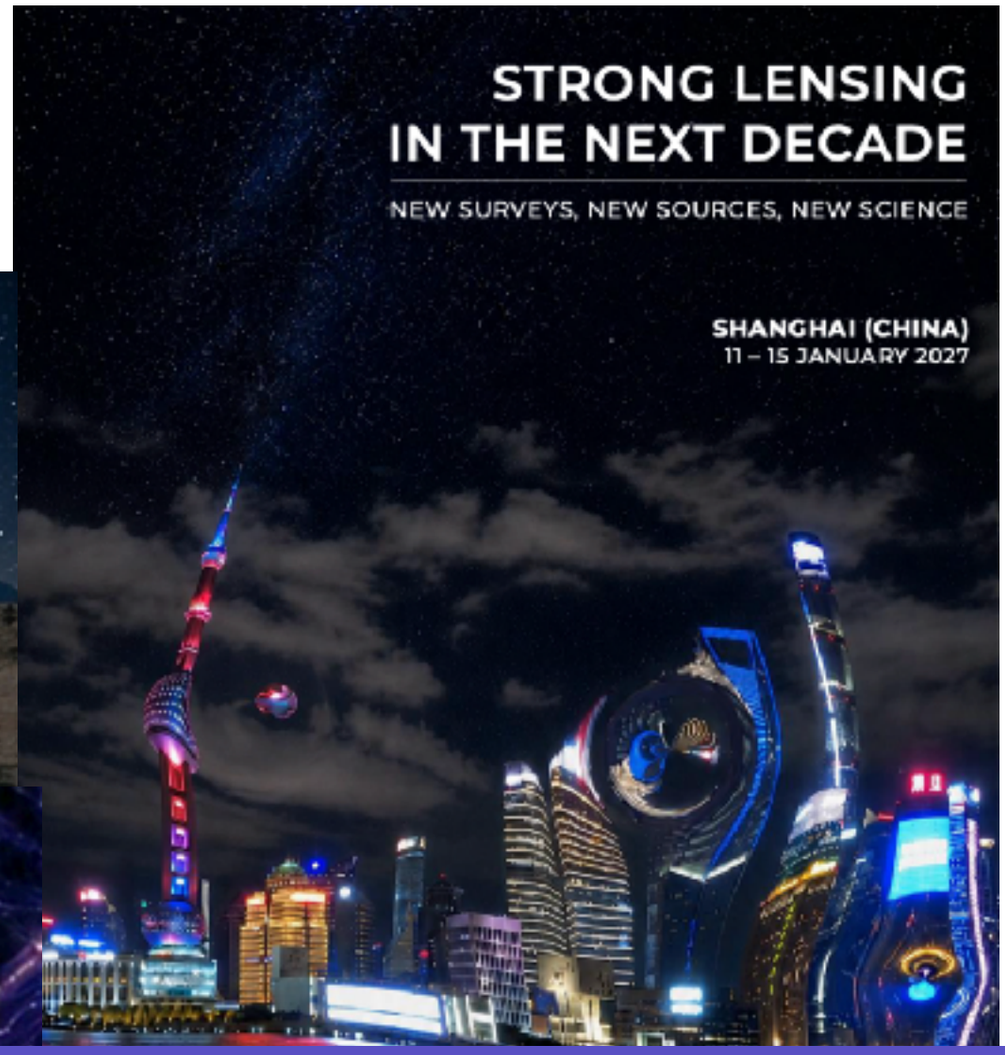
PHY 517 / AST 443:
Observational Techniques in Astronomy

Final Presentations

Going to a conference









Where to find conferences?

https://www.cadc-ccda.hia-iha.nrc-cnrc.gc.ca/en/meetings/next-year/

Goals for a conference attendance

- tell people about your awesome work!
- learn about the current hot topics in your field
- network

Conference format

- talks:
  - review / plenary talks
  - contributed talks (have to apply for these)
- posters:
  - displayed for the full duration; often dedicated poster session
  - often with “lightning talk” session
- social events (usually conference dinner)
- coffee breaks!!!

PHY517/AST443 Final presentations

- graduate students: make a poster + 1 minute "lightning talk"
- undergraduates: give a presentation; 12 minute talk + 3 minutes questions
- undergraduates who have already fulfilled the SPK requirement: can do presentation or poster

  (send me your transcript to show SPK has been fulfilled)

Final presentations

- posters will also be asked to attend the poster session of the physics graduate lab (free food + talk to many people in the department), date TBD

Final presentations

- **Mon., Dec 7th** (last day of class)

- For each presentation, you will fill out a grading rubric and assign a score (0-10). We will pass them to the presenter after anonymizing the feedback.

- The SBU astronomy group will be invited to listen in

Topics

- select one of your lab experiments
- within your group, one of you has to present a talk on Lab 3
- avoid (if possible) having two talks on the same lab within your group
- if you do research in **observational astronomy**, you can present your research instead of a Lab

Presentation structure

- Title (slide):
  - Title: be descriptive! (I.e. NOT "AST443 Final Presentation")
  - Speaker name, with affiliation
  - Co-authors
  - Venue, date
  - Good to include: affiliation logo, funding source logo (if applicable), pretty picture relevant to your talk
  - Posters: good to include picture of yourself so that people can come find you

Presentation structure

- Background / introduction
  - Present the big picture
  - Introduce the main concepts
  - Describe your target
  - Summarize previous work
  - Clearly state the question(s) your project addresses

Presentation structure

- Data / observations
  - Equipment
  - Important information depends on project, e.g.
    - Date of observations (time-variable observations)
    - Filter (imaging)
    - Grating (spectroscopy)
    - …

Presentation structure

- Data analysis and measurements
  - "Basic" data reduction does not have to explained (but can be mentioned) - by now, everybody should know what a dark frame is
  - Describe analysis choices, e.g. lightcurve binning + estimates of uncertainties
  - Describe measurements clearly, e.g. transit depth

Presentation structure

- Inferred physics and interpretation
  - E.g. ratio of planet/star size
  - Comparison to expectations / literature

Presentation structure

- Conclusion
  - Summarize the main points that you want your audience to take away
  - Can include next steps, future work, etc.

How to give a good talk

- Know your audience!
- Aim: *everyone* should get something out of your talk
  - Include enough background information
  - Avoid too much jargon
  - Avoid too many equations
  - Tell a coherent story

How to give a good talk

- Slides: *visual aids* to your *story*
  - Assume ~1-2 minutes / slide
  - Don't put too much "stuff" on one slide
  - Include relevant **pictures / figures**
  - Prefer concise keywords to full sentences (let alone paragraphs)
  - Make everything legible (e.g., axis labels)
  - Use color and font style / size to highlight points, but **Don't overDO** IT
  - Don't use yellow, light green, low-contrast colors

How to give a good talk

- Speaking:
  - Don’t speak too fast
  - Prepare not just your slides, but also what you will say
  - … but don’t memorize your talk, **speak freely**
  - *Your tone and articulation play an important part in conveying your story*
  - Engage with your audience - make eye contact
  - Avoid too many “umm”s - better to pause
  - **Practice** your talk, more than once, with different people!

How to give a good talk

- References, and avoiding plagiarism
  - Make sure to give proper credits
  - Every figure (that you did not make) needs to reference the author
  - Every research result needs to be properly cited with author / collaboration name + year; good to include journal, etc. *on the slide it is shown*
  - Visibly acknowledge your co-authors when presenting your own research, e.g. on title slide

How to make a good poster

- *Many of the same guidelines as for talks*
- Avoid too much text!!!
- Clearly structure your poster
- Make sure figures and text are well legible
- Include your picture + e-mail address

“Lightning talk” ?

- One slide, one minute - advertise your poster!

Practicalities

- You'll have to tell me your title ahead of time (for scheduling)
- Send me your talk in google slides or pdf format, well before the start of class